DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Assignment No. 4

(Trees)

Course Name & Code: Data Structure (KCS-301) Session: 2021-22 (Odd Semester)

C 203	Course Outcomes (Statements)	Bloom's Knowledge Level (KL)
C203.1	To Describe how arrays, linked lists, stacks, queues, trees, and graphs are represented in memory, used by the algorithms and their common applications.	K1, K2
C203.2	To Discuss the computational efficiency of the sorting and searching algorithms.	K2
C203.3	To Implementation of Trees and Graphs and perform various operations on these data structure.	K3
C203.4	To Understanding the concept of recursion, application of recursion and its implementation and removal of recursion.	K4
C203.5	To Identify the alternative implementations of data structures with respect to its performance to solve a real-world problem.	K5, K6

Section – A (Conceptual type Questions)

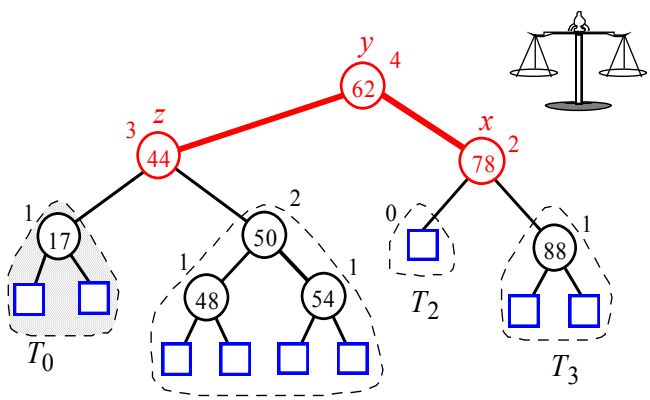
1. What is tree? What is binary tree? What do you mean by complete binary tree? Draw example of each.
2. Compare and contrast between a binary and binary search tree. Name the operations that are specific to binary search trees not to binary tree.
3. For a binary tree T, the pre-order and in-order traversal sequences are as follows:
   Pre-order: A B L M K N P Q
   In-order: L B M A N K Q P
   Draw the binary tree T.
4. For a binary tree T, the in-order and post-order traversal sequences are as follows:
   in-order: D C K E A H B Q J I
   post-order: D K E C H Q J I B A
   Draw the binary tree T.
5. How an AVL tree differs from a binary search tree? How an AVL tree are represented in computer memory?
6. What is a B-Tree? Generate a B-Tree of order 4 with the alphabets (letters) arrive in the sequence as follows:

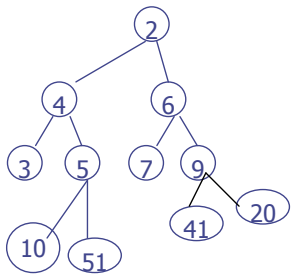
A, G, F, B, K, D, H, M, J, E, S, I, R, X, C, L, N, T, U, P

7. What is the difference between a Binary Search Tree (BST) and HEAP? For a given sequence of numbers, construct a heap and a BST.
   34, 23, 67, 45, 12, 54, 87, 43, 98, 75, 84, 93, 31
8. What is a Threaded Binary Tree? Explain the advantages of using a threaded binary tree.
9. Describe Huffman algorithm. Draw a Huffman tree and codes for the following symbols whose frequency of occurrence in a message is stated along with the symbol below :
   A : 15, B : 6, C : 7, D : 12, E : 25, F : 4, G : 6, H : 1, I : 15;
   Also Decode the message: 1110100010111011.
10. Write a C function to count the number of leaf nodes in a binary tree.

Section – B (Application based Problems)

1. Insert, into an initially empty binary search tree the following keys (in this order|): 30, 40, 24, 58, 48, 26, 11, 13. Draw the tree after each insertion.
2. Draw the AVL tree resulting from the insertion of an item with key 52 into the following AVL tree.



3. Draw the AVL tree resulting from the removal of the item with key 62 from the above AVL tree.

3. Give the preorder, postorder and inorder traversals of the nodes in the following tree.